🚀 **NASA SPACE APPS CHALLENGE 2026**

AST RAVITAL AI — 240 SECONDS OF GLORY

Official Competition Video Presentation Script

Project: AstraVital AI — Intelligent Astronaut Health Monitoring & Digital Twin Platform
Team: Team Zero
Location: Cumilla, Bangladesh
Duration: 4 Minutes (240 Seconds)

SECTION 01 — WHO: ATTENTION & AUTHENTICITY

Timestamp: 0:00 – 0:30 | 30 Seconds

Imagine an astronaut millions of kilometers from Earth. No hospital. No emergency room. And no doctor just a phone call away.

When a critical health warning appears, a distress signal to Houston takes twenty-two minutes just to arrive.

We are Team Zero, from Cumilla, Bangladesh, united by one mission for NASA Space Apps 2026: zero distance, zero delay, zero limits.

SECTION 02 — WHY: THE PROBLEM & ITS IMPORTANCE

Timestamp: 0:30 – 1:10 | 40 Seconds

Deep space travel pushes human biology to its breaking point.

In microgravity, cephalic fluid shifts trigger Spaceflight Associated Neuro-ocular Syndrome—or SANS. Cosmic radiation damages DNA, while bone density degrades over one percent every month.

NASA has spent decades studying these risks through the Human Research Program.

But on a three-year voyage to Mars, astronauts cannot carry hospital infrastructure. They need autonomous, scientifically validated health intelligence that functions completely independently of Earth.

SECTION 03 — WHAT: OUR BIG IDEA & INNOVATION

Timestamp: 1:10 – 1:55 | 45 Seconds

Our solution is AstraVital AI: an intelligent astronaut health monitoring and physiological digital twin platform.

Rather than relying on isolated algorithms, our platform is grounded directly in authentic NASA research.

We integrate the NASA Open Science Data Repository (OSDR) and GeneLab—incorporating landmark datasets like the NASA Twins Study—paired with clinical practice standards from the NASA Technical Reports Server (NTRS).

We also establish strict data provenance, transparently distinguishing between authentic NASA research, simulated telemetry for demonstration, and derived clinical analytics.

SECTION 04 — HOW: OUR LIVE WEBSITE DEMONSTRATION

Timestamp: 1:55 – 3:25 | 90 Seconds

Here is AstraVital AI in action.

On our Mission Dashboard, ten vital telemetry streams update continuously, prominently labeled as simulated data. We can adjust mission simulation seeds, trigger solar particle warnings, and track physical exercise countermeasures in real time.

When we inspect an elevated parameter—such as heart rate or intracranial pressure—our Research-to-Dashboard connection instantly displays corresponding NASA OSDR studies and NTRS publications, providing full DOI links and biological context.

In our NASA Research Explorer, users can query authentic multi-omics datasets like OSD-205 and muscle deconditioning studies. The NTRS Library provides access to NASA-STD-3001 standards and SANS clinical guidelines.

Next, our Interactive 3D Digital Twin maps microgravity organ stress across the left ventricle, optic nerve, axial spine, and femur, dynamically coupling with incoming telemetry.

Finally, our Mission Simulator projects bone resorption, cumulative radiation, and mission readiness across an entire 180 to 1,000-day Mars transit.

Our goal is not to replace flight surgeons, but to provide actionable, research-backed health intelligence.

SECTION 05 — IMPACT: FUTURE VISION & CLOSING

Timestamp: 3:25 – 4:00 | 35 Seconds

And the potential reaches far beyond space.

This same architecture can support personnel in submarines, polar research stations, disaster-response zones, and remote rural communities with limited healthcare access.

Inspired by NASA's commitment to exploration and open science, Team Zero wants to make health intelligence transparent, autonomous, and accessible.

Because exploration should never mean leaving human wellbeing behind.

TEAM ZERO BELIEVES IN ONE MOTTO:

"Zero Distance. Zero Delay. Zero Limits." 🚀